

Triangle Congruence & Proof

SUPPORTING SKILLS LEGEND

Student learning objectives for this unit

I — Introduce

D — Deepen

A — Apply

R — Reinforce

☐ 1. I can **classify** triangles by sides and angles.

RELATED SKILLS

☐ I Classify a triangle by side lengths and angle measures.☐ I Recall and correctly use terminology and notation for triangle classification, congruence, and corresponding parts.☐ A Add congruence, parallel, perpendicular, and angle markings to a diagram.

☐ 2. I can **use** triangle angle-sum and exterior-angle relationships.

RELATED SKILLS

☐ I Apply the triangle angle-sum theorem.☐ I Apply the triangle exterior-angle theorem.☐ A Use inverse operations to maintain an equivalent equation.

☐ 3. I can **write** a congruence statement with corresponding vertices in the correct order.

RELATED SKILLS

☐ I Match corresponding vertices, sides, and angles in order.☐ I Write congruence statements in corresponding order.☐ A Recall and correctly use terminology and notation for triangle classification, congruence, and corresponding parts.

☐ 4. I can **determine** whether triangles are congruent by SAS.

RELATED SKILLS

☐ I Select SAS, SSS, HL, ASA, or AAS from marked information.☐ D Select a theorem that connects known facts to a target statement.☐ A Match corresponding vertices, sides, and angles in order.

☐ 5. I can **determine** whether triangles are congruent by SSS.

RELATED SKILLS

☐ D Select SAS, SSS, HL, ASA, or AAS from marked information.☐ A Match corresponding vertices, sides, and angles in order.

☐ 6. I can **determine** whether right triangles are congruent by HL.

RELATED SKILLS

☐ D Select SAS, SSS, HL, ASA, or AAS from marked information.☐ A Use right-angle relationships created by perpendicular lines.

☐ **7. I can **determine** whether triangles are congruent by ASA or AAS.**

RELATED SKILLS

☐ D Select SAS, SSS, HL, ASA, or AAS from marked information.

☐ A Apply the triangle angle-sum theorem.

☐ **8. I can **apply** isosceles and equilateral triangle theorems and their converses.**

RELATED SKILLS

☐ I Apply isosceles-triangle theorems and converses.

☐ D Recognize and apply the converse of a conditional statement.

☐ **9. I can **use** corresponding parts of congruent triangles after proving congruence.**

RELATED SKILLS

☐ I Use corresponding parts only after triangle congruence is established.

☐ A Write congruence statements in corresponding order.

☐ **10. I can **write** a structured proof involving triangle congruence.**

RELATED SKILLS

☐ D Organize statements and reasons into a valid proof sequence.

☐ A Select SAS, SSS, HL, ASA, or AAS from marked information.

☐ A Translate givens into diagram markings and statements.

☐ A Use corresponding parts only after triangle congruence is established.
